LAB # 5
To Reproduce Device Configuration and Show Network Convergence using Static and Default Routing

Objectives

- Preparation of basic configuration on router based on CLI.
- Be able to manipulate configuration sequence in order prepare, configure and activate serial and Ethernet interfaces using Packet Tracer.
- To show connectivity between two devices using static routes.

Pre-Lab Exercise

Read this experiment in its entirety to become familiar with objectives of this lab. Also review the portions of chapter 4 of your reference book and try to understand the Routing algorithms. You may record the terms and sections that require more elaboration for reference. The instructor may provide the class some time to reflect upon these before proceeding with the lab.

Routing Basics

Once you create an internetwork by connecting your WANs and LANs to a router, you'll need to configure logical network addresses, like IP addresses, to all hosts on that internetwork for them to communicate successfully throughout it.

The term *routing* refers to taking a packet from one device and sending it through the network to another device on a different network. Routers don't really care about hosts— they only care about networks and the best path to each one of them. The logical network address of the destination host is key to get packets through a routed network. It's the hardware address of the host that's used to deliver the packet from a router and ensure it arrives at the correct destination host. List of the minimum factors a router must know to be able to affectively route packets is:

- Destination address
- Neighbor routers from which it can learn about remote networks
- Possible routes to all remote networks
- The best route to each remote network
- How to maintain and verify routing information

The router learns about remote networks from neighboring routers or from an administrator. It builds a routing table, which is basically a map of the internetwork, and it describes how to find remote networks. If a network is directly connected, then the router already knows how to get to it. You have observed in previous lab that directly connected networks are already in the routing table using *show iproute* command. The router use one of the following two ways to learn how to get to the remote network.

- Static Routing
- Dynamic Routing

Static Routing

The static routing method requires someone to hand-type all network locations into the routing table, which can be a pretty daunting task when used on all but the smallest of networks!

Dynamic Routing

In dynamic routing, a protocol on one router communicates with the same protocol running on neighboring routers. The routers then update each other about all the networks they know about and place this information into the routing table. If a change occurs in the network, the dynamic routing protocols automatically inform

all routers about the event. If static routing is used, the administrator is responsible for updating all changes by hand onto all routers.

We will discuss dynamic routing in detail in upcoming labs. In this lab we will implement static routing to access remote networks.

There are some important things you should know about routing protocols before we get deeper into them. Being familiar with administrative distances, the three different kinds of routing protocols, and routing loops are three of the most important.

Administrative Distances

The *administrative distance (AD)* is used to rate the trustworthiness of routing information received on a router from a neighbor router. An administrative distance is an integer from 0 to 255, where 0 is the most trusted and 255 means no traffic will be passed via this route.

If a router receives two updates listing the same remote network, the first thing the router checks is the AD. If one of the advertised routes has a lower AD than the other, then the route with the lowest AD will be chosen and placed in the routing table. If both advertised routes to the same network have the same AD, then routing protocol metrics like *hop count* and/or the bandwidth of the lines will be used to find the best path to the remote network. The advertised route with the lowest metric will be placed in the routing table, but if both advertised routes have the same AD as well as the same metrics, then the routing protocol will load-balance to the remote network, meaning the protocol will send data down each link.

Default Administrative Distances

Route Source	Default AD
Connected interface	0
Static route	1
EIGRP	90
OSPF	110
RIP	120

As you will progress to dynamic routing you will understand administrative distance and metrics effectively. For static routing metrics value is 0 as router didn't make any decision for route its administrator who configured the routes manually.

In-Lab Exercise

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Figure 12. Use any Class C Network ID and fill in the Addressing Table to apply an addressing scheme to the network devices. You will then perform the initial router configurations required for connectivity according to your IP assignment. After completing the basic configuration, test connectivity between the devices on the network. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected. Static routes must be configured on the routers for end-to-end communication to take place between the network hosts. You will configure the static routes that are needed to allow communication between the hosts. View the routing table after each static route is added to observe how the routing table has changed.

Topology Diagram

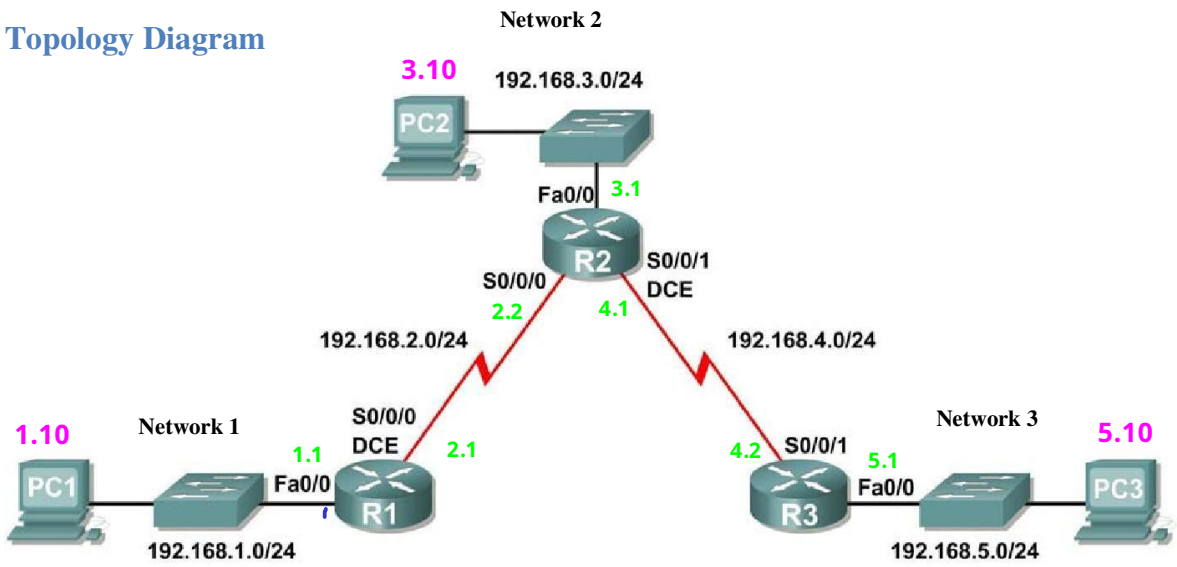

Figure 12: Network Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.2.1	255.255.225.0	N/A
R2	Fa0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/0	192.168.2.2	255.255.255.0	N/A
	S0/0/1	192.168.4.1	255.255.225.0	N/A
R3	FA0/0	192.168.5.1	255.255.255.0	N/A
	S0/0/1	192.168.4.2	255.255.255.0	N/A
PC1	NIC	192.168.1.10	255.255.225.0	192.168.1.1
PC2	NIC	192.168.3.10	255.255.255.0	192.168.3.1
PC3	NIC	192.168.5.10	255.255.255.0	192.168.5.1

Task 1: Perform Basic Router Configuration

Draw the topology diagram on Cisco Packet Tracer. Perform basic configuration on all three routers i.e. assigning IP on LAN and WAN links and other necessary administrative configurations.

Task 2: Configure IP Addressing on the Host PCs

Note: If you have difficulty with any of the commands in task 1 and task 2, see Lab 2

Task 3: Test and Verify the Configurations

Step 1: Test Connectivity of LANs

Test connectivity by pinging from each host to the default gateway that has been configured for that host.

From the host PC1, is it possible to ping the default gateway?_yes_____

From the host PC2, is it possible to ping the default gateway? yes_____

From the host PC3, is it possible to ping the default gateway? yes_____

Step 2: Test Connectivity of WANs

Check the router interfaces using the ***show ip interface brief*** command. Are all relevant interfaces **up**?_yes_____

From the router R2, is it possible to ping R1?

yes_____

From the router R2, is it possible to ping R3?

yes_____

Step 3: Routing Table

Run ***show ip route*** command on each router and answer the following.

What networks are present in the Topology Diagram but not in the routing table for R1?

_____

networks missing in R1 table: 192.168.3.0/24, 192.168.4.0/24, 192.168.5.0/24

_____

_____

_____

What networks are present in the Topology Diagram but not in the routing table for R2?

_____

networks missing in R2 table: 192.168.1.0/24, 192.168.5.0/24

_____

_____

_____

What networks are present in the Topology Diagram but not in the routing table for R3?

networks missing in R3 table: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Why are all the networks not in the routing tables for each of the routers?

routers initially know only directly connected networks

What can be added to the network so that devices that are not directly connected can ping each other?

by adding routing either static or dynamic routing

Task 4: Configure a Static Route Using a Next-Hop Address.

To configure static routes with a next-hop specified, use the following syntax:

Router(config)# **iproute***network-address subnet-mask ip-address*

- *network-address*—Destination network address of the remote network to be added to the routing table.
- *subnet-mask*—Subnet mask of the remote network to be added to the routing table. The subnet mask can be modified to summarize a group of networks.
- *ip-address*—Commonly referred to as the next-hop router's IP address.

Step 1: On the R3 router, configure a static route to the network 2 using the Serial 0/0/1 interface of R2 as the next-hop address. Complete the following command and run it on R3.

R3(config)#**ip route** 192.168.3.0 255.255.255.0 192.168.4.1

Step 2: On the R2 router, configure a static route to reach the network 3using the Serial 0/0/1 interface of R3 as the next-hop address. Complete the following command and run it on R2.

R2(config)#**ip route** 192.168.5.0 255.255.255.0 192.168.4.2

Step 3: On the R2 router, configure a static route to reach the network 1 using the Serial 0/0/0 interface of R1 as the next-hop address. Complete the following command and run it on R2.

R2(config)#**ip route** ip route 192.168.1.0 255.255.255.0 192.168.2.1

Step 4: On the R1 router, configure a static route to the network 2using the Serial 0/0/0 interface of R2 as the next-hop address. Complete the following command and run it on R1.

R1(config)#**ip route** ip route 192.168.3.0 255.255.255.0 192.168.2.2

Task 5: Test and Verify the Static Routes

Step 1: View the routing table to verify the new static route entry

Run ***show ip r*oute** command on each router write down your observations.

new entries appear with S code in show ip route

Step 2: Use ping to check connectivity.

Ping to check connectivity between the host PC2 and PC1, is ping successful and why?

yes. R2 has route to 192.168.1.0/24, R1 has route to 192.168.3.0/24

Ping to check connectivity between the host PC3 and PC2, is ping successful and why?

yes. R3 has route to 192.168.3.0/24, R2 has route to 192.168.5.0/24

Ping to check connectivity between the host PC1 and PC3, is ping successful and why?

no, full end-to-end routes are not complete yet

Task 6: Configure a Default Static Route

In the previous steps, you configured the router for specific destination routes. But could you do this for every route on the Internet? No. The router and you would be overwhelmed. To minimize the size of the routing tables, add a default static route. A router uses the default static route when there is not a better, more specific route to a destination.

Instead of filling the routing table of R1 with static routes, we could assume that R1 is a *stub router*. This means that R2 is the default gateway for R1. If R1 has packets to route that do not belong to any of R1 directly connected networks, R1 should send the packet to R2. However, we must explicitly configure R1 with a default route before it will send packets with unknown destinations to R2. Otherwise, R1 discards packets with unknown destinations.

To configure a default static route, use the following syntax:

Router(config)#***ip route*****0.0.0.0 0.0.0.0** { *ip-address* | *interface* }

Step 1: Configure the R1 router with a default route.

Configure the R1 router with a default route using the Serial 0/0/0 interface of R1 as the next-hop interface. Run the following command on R1

R1(config)#**ip route** ip route 0.0.0.0 0.0.0.0 192.168.2.2 Or ip route 0.0.0.0 0.0.0.0 se 0/0/0

View the routing table to verify the new static route entry.

S* 0.0.0.0/0 [1/0] via 192.168.2.2

Step 2: Remove static route on R1

As we have set the default route on R1 we don't need any static route. Remove the static routes that is currently configured on R1 in Task 4 Step 4 by using the ***no*** form of the command.

Note: *If this route is not to be deleted it cause no problem to routing. It is just done to show you defaultrouting.*

R1 ~~R3~~(config)#**no ip route** 192.168.3.0 255.255.255.0 192.168.2.2

View the routing table to verify the static route entry is deleted.

yes, it was deleted

Ping to check connectivity between the host PC2 and PC1, is ping successful and why?

yes, because R1 forwards unknown destinations to R2 via default route.

Step 3: Configure the R3 router with a default route.

Configure the R3 router with a default route using the Serial 0/0/0 interface of R1 as the next-hop interface. Run the following command on R1

R3(config)#**ip route** **0.0.0.0 0.0.0.0 s0/0/1 Or 0.0.0.0 0.0.0.0 192.168.4.1**

View the routing table to verify the new static route entry.

S 192.168.1.0/24 [1/0] via 192.168.4.1

S 192.168.2.0/24 [1/0] via 192.168.4.1
S 192.168.3.0/24 [1/0] via 192.168.4.1

C 192.168.4.0/24 is directly connected, Serial3/0
C 192.168.5.0/24 is directly connected, FastEthernet0/0

S* 0.0.0.0/0 [1/0] via 192.168.4.1 <= new static route

Step 4: Use ping to check connectivity.

Ping to check connectivity between the host PC2 and PC1, is ping successful.

yes, the ping is sucessfull.

Ping statistics for 192.168.1.0:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Ping to check connectivity between the host PC3 and PC2, is ping successful.

yes, the ping is sucessfull.

Ping statistics for 192.168.3.0:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Ping to check connectivity between the host PC1 and PC3, is ping successful.

yes, the ping is sucessfull.

Ping statistics for 192.168.5.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)

Task 7: Summary, Reflection, and Documentation

With the completion of this lab, you have:

- Configured your first network with a combination of static and default routing to provide full connectivity to all networks
- Observed how a route is installed in the routing table when you correctly configure and activate and interface

- Learned how to statically configure routes to destinations that are not directly connected
- Learned how to configure a default route that is used to forward packets to unknown destinations

Finally, you should document your network implementation. On each router, capture the following command output to a text (.txt) file and save for future reference.

- **show running-config**
- **show ip route**
- **show ip interface brief**

Rubric for Lab Assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed assigned tasks without any help from the instructor and showed the results appropriately.	4	
Good	The student completed assigned tasks with minimal help from the instructor and showed the results appropriately.	3	
Average	The student could not complete all assigned tasks and showed partial results.	2	
Worst	The student did not complete assigned tasks.	1	

Instructor Signature: _____ Date: _____